Diabetic Retinopathy Grading from Fundus Photographs: Development on APTOS 2019 and External Evaluation on IDRiD

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Background. Diabetic retinopathy (DR) is a leading cause of preventable blindness, and screening relies on grading colour fundus photographs. Automated grading is promising, but models that perform well on their development data may fail on images from another clinic or camera.

Methods. APTOS 2019 (3,662 images) was used for development and IDRiD (516 original images) only for external evaluation. Both datasets were validated for integrity, duplicate photographs and conflicting labels before use. Images were cropped to the retina (Otsu threshold with a validated fallback), padded and resized with anti-aliasing. A hand-crafted radiomics baseline (43 features, ridge logistic regression) was compared with an ImageNet-pretrained ResNet18, with and without CLAHE contrast enhancement. All choices were made on a validation set under a protocol recorded before training; the final model was a 5-seed ensemble, evaluated once on an internal test set and once on IDRiD with bootstrap confidence intervals, followed by robustness tests, calibration, error analysis and Grad-CAM.

Results. Validation removed 176 APTOS images (37 duplicate groups had conflicting grades) and 1,041 IDRiD images (1,032 rotated copies, 9 duplicates). On the internal test set the ensemble reached a quadratic weighted kappa (QWK) of 0.916 (95% CI 0.892–0.936), versus 0.683 for the baseline (paired difference +0.233, 0.181–0.290), with referable-DR AUROC 0.984 and good calibration (ECE 0.025). CLAHE gave no benefit. Performance was stable across seeds (validation QWK 0.914 ± 0.003). Robustness tests showed that SSIM did not predict the damage caused by degradations. On IDRiD, QWK fell to 0.696 (0.641–0.744) and referable sensitivity to 0.62; re-weighting the internal test to IDRiD's case mix did not explain the drop (QWK 0.901). Exploratory local recalibration restored calibration but not the full discrimination.

Conclusion. A carefully validated pipeline produced a strong and stable internal result, but the model did not transfer to a new camera and site without recalibration and missed 23 of 152 severe or proliferative eyes externally. External, site-specific validation is essential before any clinical use.

Keywords: diabetic retinopathy · fundus photography · data validation · preprocessing · transfer learning · radiomics · robustness · external validation · calibration · Grad-CAM

How this project addresses the evaluation criteria

The course project is assessed on implementation rather than predictive performance. Table 1 maps each criterion to what was done and where the evidence is.

Table 1. Course criteria and how the project meets them.

Criterion	What was implemented	Section
Correct image preprocessing	Retina mask by Otsu thresholding with a fallback triggered by an overlap check (Otsu failed on 17% of APTOS images); crop and pad to square without distortion; anti-aliased area resizing, with aliasing measured on 50 paired images; black background standardised; lossless storage; CLAHE tested as an alternative and rejected on evidence.	2.4, 3.2

Criterion	What was implemented	Section
Performance evaluation strategy	Data validation first (duplicates, label conflicts, cross-dataset overlap); stratified 70/15/15 split; protocol and decision rules time-stamped before results; all choices on validation; internal test and IDRiD evaluated once; ordinal (QWK) and clinical (referable DR) metrics; bootstrap and paired-bootstrap CIs; effect sizes instead of p-values; calibration; case-mix standardisation.	2.5–2.7, 3.3–3.4
Robustness checks	14 controlled degradations (noise, blur, gamma, JPEG, denoising) with SSIM/MSE against change in QWK; external validation on another camera and site; prevalence-independent per-grade recall; performance by image-quality tertile; Grad-CAM shortcut check on all 1,030 test images.	3.6–3.8
Tricks to improve stability	Transfer learning; class-weighted loss; augmentation; warm-up and cosine learning rate; early stopping on validation QWK; 5-seed ensemble (better than every single seed on test macro-F1); stability quantified across seeds.	2.6, 3.5
Especially clever implementation	Duplicate threshold taken from a gap in the similarity distribution; task-specific vessel contrast (black-hat Weber) to choose the colour channel; case-mix standardisation to separate prevalence from domain shift; cross-fitted recalibration; rule-based (not hand-picked) example selection; resumable training; shared, versioned code modules.	2.2–2.7, 3.7
Radiomics, hand-crafted tools, neural networks	All three: GLCM/Haralick radiomics baseline; classical processing (Otsu, morphology, CLAHE, linear filters); ResNet18 CNN.	2.4, 2.6

1 Introduction

Diabetic retinopathy damages the small retinal vessels and progresses from microaneurysms and haemorrhages to new-vessel formation (proliferative DR). It is graded on the International Clinical DR scale from 0 (no DR) to 4 (proliferative); grades 2 and above are usually considered *referable* to an ophthalmologist [1]. Screening programmes photograph the retina with a fundus camera and rely on trained graders, which motivates automated grading [2, 3].

The imaging modality. A fundus photograph is an optical, non-ionising, two-dimensional colour *projection* image of the curved retina, acquired with a white-light flash and an RGB sensor. Unlike CT, PET or MRI it involves no tomographic reconstruction and no ionising dose; its main sources of variability are camera optics, field of view, illumination, focus and compression. These are exactly the factors that can make a model trained at one site fail at another.

Aim and research questions. The project asks whether a model developed on APTOS 2019 generalises to IDRiD, a dataset from a different camera and site. The primary task is 5-class grading; the secondary task is referable DR (grade ≥ 2). The report answers seven questions:

1. How trustworthy are the data, and what had to be removed?
2. Which preprocessing is justified by the images themselves?
3. Was the experiment fair, i.e. free of leakage and of decisions made on test data?
4. Does a CNN improve on a hand-crafted radiomics baseline, and does CLAHE help?
5. Are the results stable, and how robust are they to image degradation?
6. Does the model generalise to IDRiD, and if not, why and how much can recalibration recover?
7. What do the model's errors and Grad-CAM explanations reveal?

2 Materials and Methods

2.1 Datasets

APTOS 2019 (Kaggle; Aravind Eye Hospital, India) provides 3,662 labelled PNG images from several clinics and cameras, with 17 different resolutions (474–4,288 px wide). **IDRiD** [4] provides fundus images from a single camera (4,288 × 2,848 px, JPEG). The copy used here was a third-party repackaged archive of 1,548 images: exactly three copies of each of the 516 official disease-grading images (one original and two rotated copies). Both datasets

use the International Clinical DR scale. Neither provides patient identifiers. IDRiD was used *only* for external evaluation.

2.2 Data validation

Integrity. Every label was matched to an image and every image decoded (APTOS: 0 missing, 0 corrupt, all RGB). **Recovering the IDRiD originals.** The 516 images at the official camera size matched the official grade counts exactly in both the training and test parts (413 + 103), so the 1,032 images at 4,880 × 3,862 px were identified as rotated copies and removed.

Duplicate photographs. A 64-bit perceptual hash proved unusable on fundus images (1,360 false "duplicate" pairs among different eyes, because all fundus photographs share the same layout). Instead, each image was reduced to a retina-cropped, z-normalised 64 × 64 green-channel fingerprint, and all pairs were compared by correlation. The distribution of nearest-neighbour correlations showed a clear gap between about 0.994 (different eyes) and 1.000 (copies), so a threshold of $r \geq 0.995$ was used. Candidates were confirmed at higher resolution (mean absolute pixel difference 0–0.3 grey levels; byte-identical files for IDRiD) and by visual inspection of the pairs just above and just below the threshold. **Decision rule**, fixed before evaluation: keep one image of a group with the same grade; remove the whole group if grades conflict, because its label cannot be trusted. The same screening was run between the datasets.

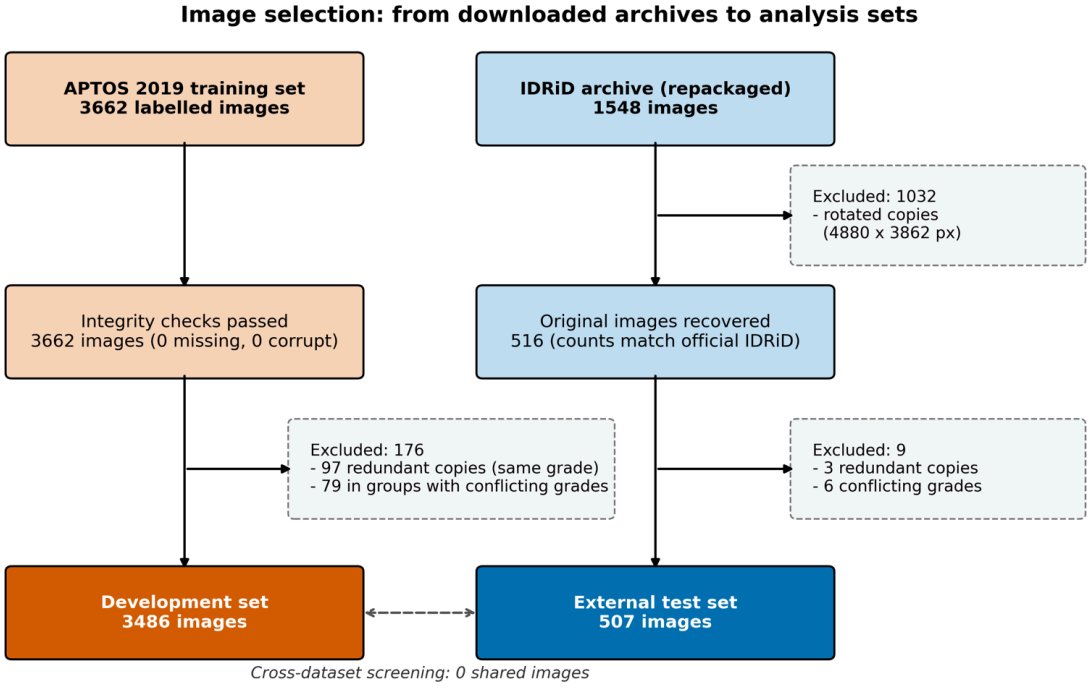


Figure 1. Image selection from the downloaded archives to the analysis sets. Exclusion counts are computed from the saved decision tables.

2.3 Exploratory image analysis

To ground preprocessing in measurements rather than assumptions, every image was standardised (retina 512 px across) and described by brightness, global contrast, sharpness (variance of the Laplacian), noise (wavelet estimate of σ) and field-of-view size. To decide which colour channel carries the diagnostic structure, a *task-specific* measure was used in addition to generic contrast: the morphological **black-hat transform** (closing minus image) isolates thin dark structures such as vessels and haemorrhages, and dividing by the local background gives a **Weber contrast**. Differences between the datasets are reported as Cliff's δ (-1 to $+1$), an effect size, rather than p-values, because with thousands of images almost any difference is "significant".

2.4 Preprocessing

Each step is justified by a property of the images:

1. **Retina mask.** Otsu's threshold on the smoothed red channel, which chooses the grey level maximising the between-class variance [5], followed by morphological opening and closing and selection of the largest filled region. Because Otsu assumes a bimodal histogram, its mask was compared with a fixed-threshold mask; if their overlap (IoU) was below 0.90, the fixed mask was used and the image was flagged for inspection.
2. **Crop and pad.** Crop to the retina and pad to a square with black, so the retina is never stretched.
3. **Anti-aliased resize to 512 × 512 px** using area interpolation, which averages pixel blocks (a low-pass filter) before sampling, as the sampling theorem requires.
4. **Background** set to zero outside the mask, so both datasets share an identical background; images stored as lossless PNG.
5. **Version B** (fixed before any training): version A plus CLAHE [6] on the lightness channel of LAB colour space (clip limit 2.0, 8 × 8 tiles). A high-pass (Ben Graham) filter was shown for illustration only.

Training augmentation (training set only): horizontal and vertical flips, rotation up to ±180°, ±5% zoom, ±10% brightness and contrast.

2.5 Experimental design

The 3,486 validated APTOS images were split, stratified by grade, into training (2,440), validation (523) and internal test (523) sets with a fixed seed; IDRiD (507) formed the external set. A protocol file recording the split, metrics, selection rule and evaluation order was saved *before* any model was trained, and a tie-break rule for the CLAHE comparison was added, time-stamped, before version B was trained (Appendix B). The order of evaluation was: compare on validation → freeze → evaluate once on the internal test → evaluate once on IDRiD. IDRiD outcomes were never used for any decision.

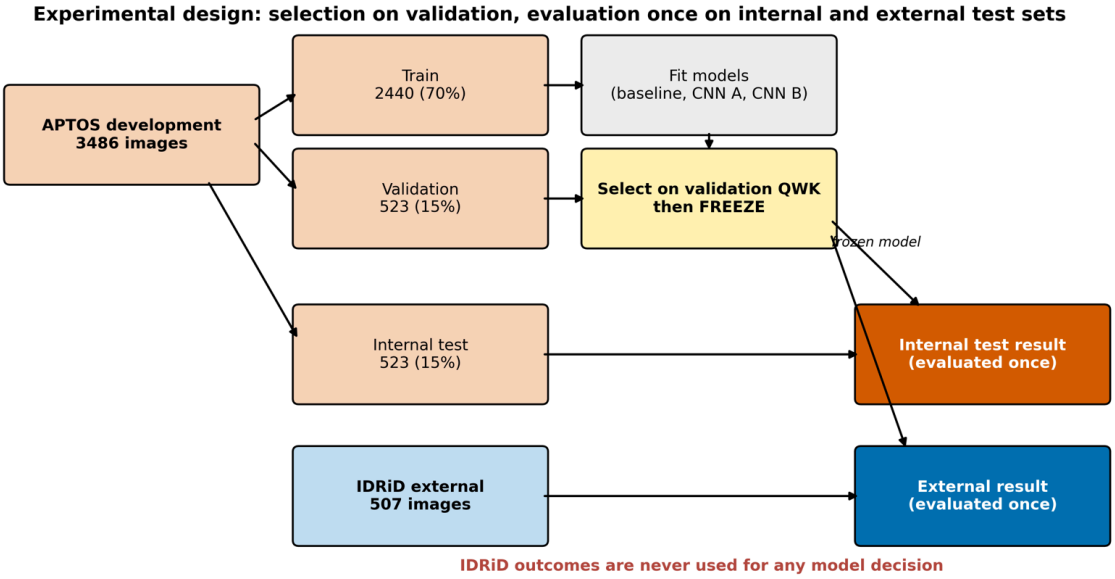


Figure 2. Experimental design. All model choices are made on the validation set; the frozen model is evaluated once on each test set.

2.6 Models

Radiomics baseline. Forty-three hand-crafted features were computed inside the retina of each version-A image: intensity statistics of the three channels (mean, SD, skewness, kurtosis, percentiles); grey-level co-occurrence (GLCM) texture features of the green channel after discretisation to 32 levels (contrast, dissimilarity, homogeneity, energy, correlation, angular second moment at distances 1, 2 and 4 px, averaged over four angles) [7]; and morphological lesion features (black-hat and top-hat responses). A standardised multinomial logistic regression

with L2 (ridge) penalty and class weights was fitted; the penalty C was chosen by 5-fold cross-validation on the training set only.

CNN. A ResNet18 [8] pretrained on ImageNet had its final layer replaced by a 5-class output and was fine-tuned on 512-px images with class-weighted cross-entropy (weights inversely proportional to training-set frequency), AdamW (learning rate 2×10^{-4}, weight decay 10^{-4}), one warm-up epoch followed by cosine decay, batch size 32, mixed precision, up to 25 epochs, and early stopping on validation QWK (patience 7), keeping the weights of the best epoch.

Stability. The frozen configuration was trained with five seeds (42, 1, 2, 3, 4) that change initialisation, augmentation and batch order. The final model is the **ensemble**: the average of the five seeds' predicted probabilities. Its referable threshold was chosen on validation to reach sensitivity ≥ 0.90 (threshold 0.653) and then frozen.

2.7 Evaluation

Metrics. The primary metric is the **quadratic weighted kappa** (QWK), suited to ordered grades because it penalises distant errors more than near misses. Secondary metrics are macro-F1, accuracy, per-grade recall, and, for referable DR, AUROC, sensitivity and specificity at the frozen threshold. **Uncertainty.** 95% confidence intervals come from 2,000 bootstrap resamples of the test images; two methods evaluated on the same images are compared with a *paired* bootstrap of their difference. Seed-to-seed variability is reported separately. **Calibration** of P(referable) is assessed with the Brier score, the expected calibration error (ECE) and a reliability diagram.

Robustness. Fifteen conditions, fixed in advance, were applied to the internal test images: Gaussian noise ($\sigma = 5$, 10, 20), Gaussian blur ($\sigma = 1$, 2, 4 px), gamma (0.6, 0.8, 1.25, 1.6), JPEG compression (quality 90, 50, 20), and noise $\sigma = 20$ followed by a 3×3 median filter. For each, image change (SSIM [9], MSE, inside the retina) was compared with model change (ΔQWK with paired-bootstrap CI, and the share of images whose predicted grade changed).

External evaluation. Besides the raw internal-versus-external comparison, the internal test set was **re-sampled to IDRiD's grade distribution** (2,000 stratified resamples). If the external result matched this case-mix-standardised internal result, the difference would be explained by prevalence; otherwise, by a change in model behaviour. Per-grade recall, which does not depend on prevalence, and performance by image-quality tertile were also reported. **Exploratory analyses**, labelled as post-hoc and performed after the frozen external evaluation, examined the shift in predicted probabilities and estimated what local recalibration could achieve by **2-fold cross-fitting** on IDRiD (threshold and Platt scaling fitted on one half, evaluated on the other).

Explainability. Grad-CAM [10] maps (last convolutional block, 16×16) were averaged over the five seeds. Eight cases were selected by a fixed rule (per test set: the two most confident correct referable cases and the two largest errors). A shortcut check on all 1,030 test images measured the share of Grad-CAM mass on the background and its concentration on the outer rim of the retina relative to the rim's area.

3 Results

3.1 How trustworthy are the data?

Tested Integrity, duplicate photographs within and between datasets, and label consistency.

Why Duplicates across splits leak information and inflate results; contradictory labels add noise to both training and evaluation.

Found APTOS: 231 candidate pairs ($r \geq 0.99$), of which 148 lay above the threshold, forming 131 duplicate groups (270 images); **37 groups (28%) had conflicting grades**, one photograph being labelled 2, 3 and 4. IDRiD: 6 byte-identical pairs, 3 with conflicting grades (1 vs 2, 4 vs 3, 0 vs 3). No image was shared between datasets (highest correlation 0.948).

Meaning 176 APTOS images (97 redundant copies, 79 in conflicting groups) and 9 IDRiD images were removed, leaving 3,486 and 507. The method detects copies of the same photograph, not different photographs of the same eye, so patient independence remains unverified.

Table 2. Analysis sets after validation (counts and percentages by grade).

Set	No DR	Mild	Moderate	Severe	Proliferative	Total	Referable
Train (APTOS)	1,257 (51.5%)	236 (9.7%)	645 (26.4%)	117 (4.8%)	185 (7.6%)	2,440	38.8%
Validation (APTOS)	269 (51.4%)	51 (9.8%)	138 (26.4%)	25 (4.8%)	40 (7.6%)	523	38.8%
Internal test (APTOS)	270 (51.6%)	51 (9.8%)	138 (26.4%)	25 (4.8%)	39 (7.5%)	523	38.6%
External (IDRiD)	167 (32.9%)	23 (4.5%)	165 (32.5%)	91 (17.9%)	61 (12.0%)	507	62.5%

The stratified split reproduces the APTOS grade mix in every subset; IDRiD has a markedly more severe case mix, which must be considered whenever internal and external results are compared.

3.2 Which preprocessing is justified by the images?

Tested Which colour channel carries the diagnostic structure; how the datasets differ; whether the retina mask works; whether resizing needs anti-aliasing; whether CLAHE harmonises the datasets.

Why Preprocessing should follow from measured properties of the data, not from habit.

Found A generic contrast measure favoured the red channel, but the task-specific vessel Weber contrast favoured **green by a factor of about 2.3** in both datasets (0.156 vs 0.069 APTOS; 0.168 vs 0.064 IDRiD). The datasets differed most in field-of-view size (Cliff's $\delta = 0.97$) and noise ($\delta = 0.76$), and hardly in sharpness ($\delta = 0.08$). Otsu's mask failed on **596 APTOS images (17%)**, almost all underexposed, where the histogram is not bimodal and Otsu separated the bright optic disc from the rest; the fallback produced complete masks (smallest retina fraction 0.63). Without a low-pass filter, downsampling raised the estimated noise **2.7-fold** and the Laplacian variance **5.5-fold** in 50/50 paired images. CLAHE did not reduce the differences between datasets (δ for brightness 0.28 → 0.35).

Meaning Green-channel texture features were used in the baseline; area interpolation was justified quantitatively; the mask fallback was necessary. These findings also illustrate that image quality is task-specific.

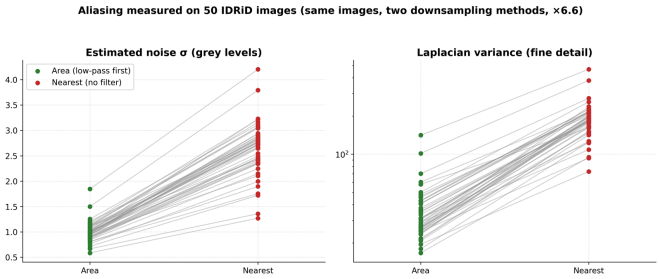


Figure 3. Aliasing measured on 50 IDRiD images reduced 6.6-fold: nearest-neighbour sampling (no low-pass filter) vs area interpolation, same images.

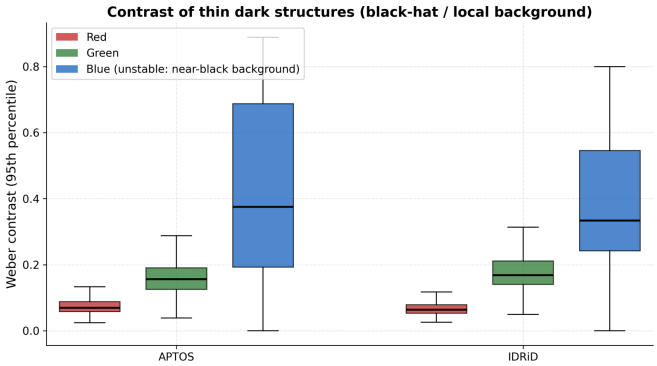


Figure 4. Contrast of thin dark structures per channel. The blue channel's high values are unstable because its background is near-black.

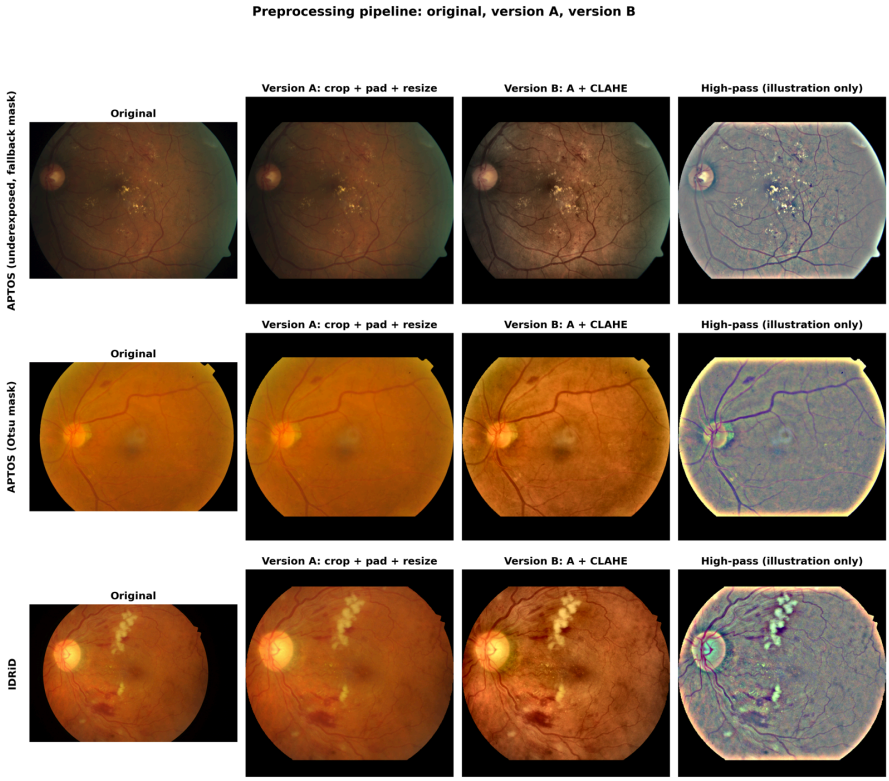


Figure 5. Preprocessing pipeline on three images chosen by a fixed rule: original, version A, version B (CLAHE), and a high-pass filter shown for illustration only (note its artificial bright rim, an edge effect of linear filtering).

3.3 Does the CNN improve on the baseline, and does CLAHE help? (validation)

Tested Radiomics baseline vs CNN-A vs CNN-B on the validation set, with paired-bootstrap differences.

Why Selection must happen on validation so the test sets remain untouched.

Found The baseline's cross-validated QWK rose with C and plateaued for C ≥ 100 (0.741 vs 0.742), so C = 1000 was used. CNN-A exceeded the baseline by **+0.219 QWK** (95% CI 0.167–0.276). CNN-B minus CNN-A was **−0.003** (−0.025 to +0.017); the CI includes 0.

Meaning Deep learning clearly beats hand-crafted features. CLAHE visibly enhances vessels but gives the CNN no measurable benefit, so by the rule recorded before CNN-B was trained, the simpler version A was frozen. A plausible explanation is that early convolutional layers learn local contrast normalisation themselves.

Table 3. Validation results (n = 523) used for selection.

Model	QWK	Macro-F1	Accuracy	Referable AUROC	Specificity at 90% sensitivity
Radiomics baseline	0.700	0.498	0.673	0.931	0.831
CNN-A (standard)	0.919	0.733	0.855	0.987	0.975
CNN-B (CLAHE)	0.916	0.726	0.858	0.984	0.962

The learning curves showed validation loss reaching its minimum around epoch 11 and then rising while validation QWK continued to improve slightly. Loss scores the probabilities and penalises confident mistakes heavily, whereas QWK scores only the predicted grade: the model was becoming overconfident while its decisions stayed mostly correct.

3.4 Main result: the internal test set (evaluated once)

Tested The frozen 5-seed ensemble on the 523 internal test images, with the threshold fixed on validation.

Why This is the unbiased estimate of performance on images from the development source.

Found QWK **0.916** (0.892–0.936), almost identical to validation (0.918), so epoch selection did not inflate the result. Referable AUROC 0.984; sensitivity 0.866 (0.817–0.912) and specificity 0.953 (0.930–0.975). The ensemble beat the baseline by +0.233 QWK (0.181–0.290) and +0.073 AUROC (0.051–0.097). Calibration was good: Brier 0.051 and ECE 0.025 (baseline 0.120 and 0.052).

Meaning The pipeline works well on its own source. Two caveats: the threshold chosen for 90% sensitivity gave 86.6% on new images, and recall for the rare grades was low and uncertain (proliferative 0.49, 0.32–0.65).

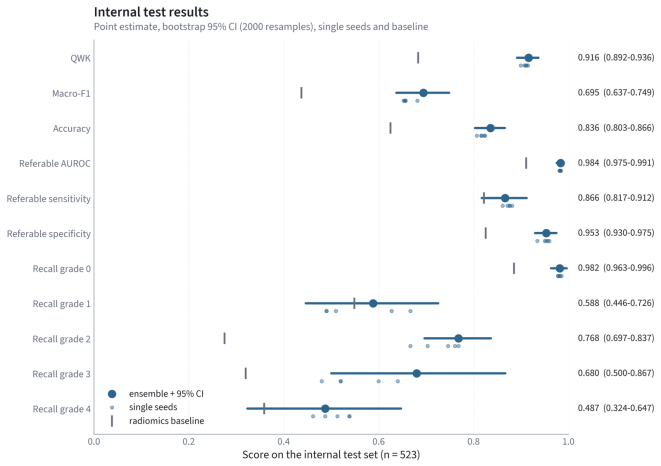


Figure 6. Internal test results: ensemble with bootstrap 95% CI, single seeds and the radiomics baseline.

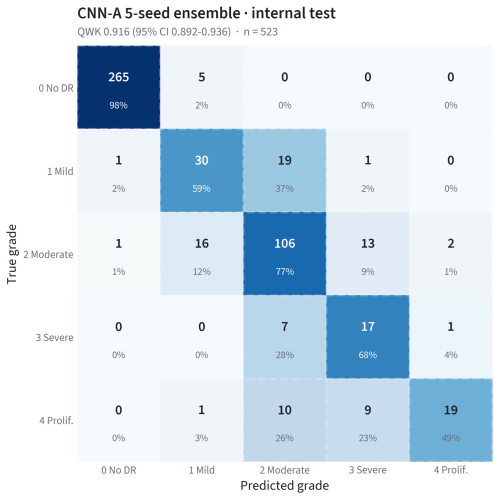


Figure 7. Internal test confusion matrix: errors are mostly between neighbouring grades.

3.5 Are the results stable?

Tested The frozen configuration retrained with five seeds; per-seed metrics versus the ensemble.

Why A single run may be lucky; averaging seeds is a standard stability technique.

Found Validation QWK 0.914 ± 0.003 (0.911–0.919); seed 42, the one used for selection, was the best of the five. On the internal test, single seeds reached QWK 0.909 ± 0.006 and macro-F1 0.660 ± 0.012, while the ensemble reached 0.916 and **0.695, higher than every single seed**. Rare-grade recall varied much more (validation proliferative recall 0.43–0.63).

Meaning The overall result does not depend on a lucky seed, and ensembling improves both accuracy and calibration. Per-grade results for rare classes remain uncertain because there are few such images.

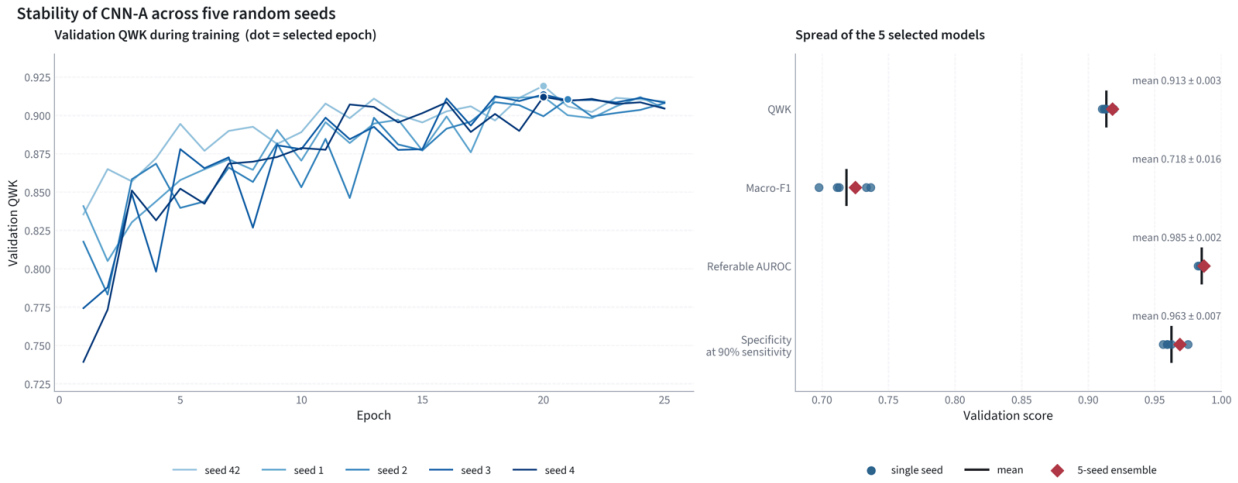


Figure 8. Validation QWK during training for five seeds (dot = selected epoch) and the spread of the selected models; diamond = 5-seed ensemble.

3.6 How robust is the model to image degradation?

Tested Fifteen degradation conditions applied to the internal test images; image change (SSIM, MSE) vs model change (ΔQWK, share of changed grades).

Why Real screening images vary in noise, focus, exposure and compression, and a generic quality metric may not predict diagnostic harm.

Found **SSIM did not predict the damage.** Darkening ($\gamma = 1.6$, SSIM 0.71) was harmless (ΔQWK −0.015), whereas blur $\sigma = 4$ (SSIM 0.90) cost −0.49 and JPEG quality 20 (SSIM 0.86) cost **−0.82**, changing 82% of predicted grades. Errors went in opposite clinical directions: strong noise made the model *miss* disease (sensitivity 0.26), strong compression made it *over-call* disease (specificity 0.38). Noise was harmful even at $\sigma = 5$ (−0.067). Brightening hurt more than darkening ($\gamma = 0.6$: −0.18). A median filter after strong noise raised SSIM from 0.20 to 0.56 but QWK only from 0.36 to 0.40.

Meaning Image quality must be judged by its effect on the task, not by generic similarity. The model is fragile to noise and strong compression, which motivates noise augmentation and image-quality control in deployment.

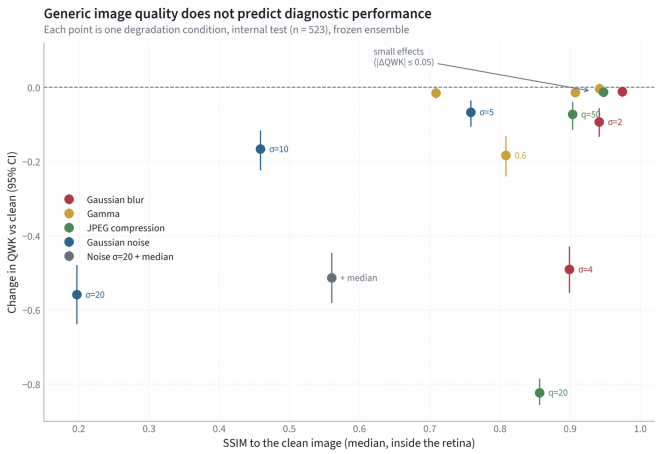


Figure 9. SSIM to the clean image vs change in QWK (95% CI) for each condition.

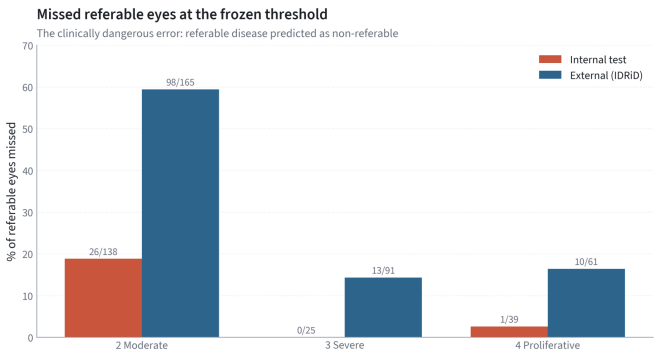


Figure 10. Referable eyes missed at the frozen threshold, internal vs external (see 3.7).

3.7 Does the model generalise to IDRiD?

Tested The frozen ensemble on the 507 IDRiD images (evaluated once), compared with the internal test raw and after re-weighting to IDRiD's case mix; per-grade recall; calibration; missed referable eyes.

Why A validation score from the development source cannot show performance with a new camera, population and graders.

Found External QWK **0.696** (0.641–0.744), a difference of −0.220 (−0.282 to −0.166). Re-weighting the internal test to IDRiD's case mix lowered QWK only to 0.901, so **case mix does not explain the drop**. Referable AUROC stayed fairly high (0.937), but at the frozen threshold sensitivity fell to 0.62 while specificity rose to 0.99, and calibration failed (ECE 0.20 vs 0.025). Predictions were compressed towards the middle grades (mean predicted − true grade: +0.50 for No DR, −0.52 for Moderate, −0.68 for Severe, −1.21 for Proliferative). At the frozen threshold the model missed 98/165 moderate, 13/91 severe and **10/61 proliferative** eyes. Within IDRiD, performance showed no consistent trend across noise or sharpness tertiles. Seeds disagreed more externally (QWK SD 0.020 vs 0.006). The baseline degraded more (AUROC 0.911 → 0.769).

Meaning The model's ranking of eyes partly transfers, but its probabilities and threshold do not. The cause is a dataset-level change in behaviour, not prevalence; this design cannot separate camera, illumination and grading-convention effects.

Table 4. Internal vs external performance of the frozen ensemble (95% bootstrap CI).

Metric	Internal test	Internal at IDRiD case mix	External (IDRiD)	External − internal
QWK	0.916 (0.892–0.937)	0.901 (0.877–0.920)	0.696 (0.641–0.744)	−0.220 (−0.282 to −0.166)
Macro-F1	0.695 (0.637–0.749)	0.693 (0.649–0.740)	0.401 (0.357–0.444)	−0.293 (−0.362 to −0.220)
Referable AUROC	0.984 (0.975–0.991)	0.990 (0.984–0.994)	0.937 (0.915–0.957)	−0.046 (−0.069 to −0.025)
Referable sensitivity	0.866 (0.817–0.912)	0.896 (0.865–0.927)	0.618 (0.563–0.671)	−0.248 (−0.319 to −0.175)
Referable specificity	0.953 (0.930–0.975)	0.964 (0.944–0.985)	0.990 (0.973–1.000)	+0.036 (0.010 to 0.063)
Recall, Severe (grade 3)	0.680 (0.500–0.867)	0.681 (0.585–0.766)	0.187 (0.109–0.268)	−0.493 (−0.697 to −0.285)

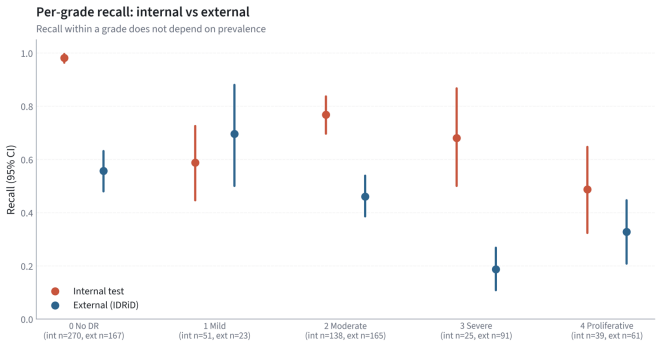


Figure 11. Per-grade recall, internal vs external; recall within a grade does not depend on prevalence.

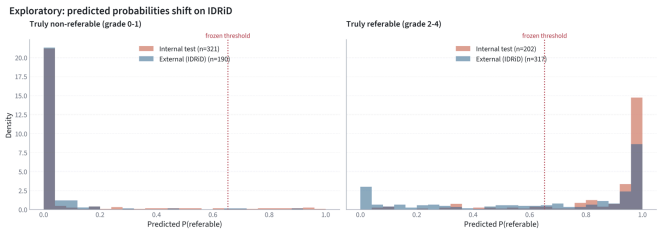


Figure 12. Exploratory: P(referable) by true status. On IDRiD many referable eyes receive low probabilities, some close to 0.

Exploratory: how much can local recalibration recover? With 2-fold cross-fitting on IDRiD, a locally chosen threshold restored sensitivity to 0.905 but with specificity 0.784 (internally 0.953), and Platt scaling reduced the ECE from 0.201 to 0.046 and the Brier score from 0.163 to 0.095. The shift therefore has two parts: a **calibration shift** that local recalibration can fix, and a **loss of discrimination** that it cannot. This mirrors the standard sequence for clinical prediction models: when external validation fails, recalibrate and then validate again on independent data.

3.8 What do the errors and explanations reveal?

Tested Error magnitude; image quality of large errors vs correct predictions; Grad-CAM on rule-selected cases; a shortcut check on all test images.

Why To understand failure modes and check whether the model relies on artefacts such as the image border.

Found Internally 83.6% of grades were exact and 2.9% off by two or more; on IDRiD 43.8% exact, 47.7% off by one and 8.5% off by two or more. Internally, large errors had lower sharpness (Cliff's $\delta = -0.45$) and contrast ($\delta = -0.34$), on only 15 cases. Grad-CAM mass on the outer rim was 0.57–0.61 times what its area would suggest, and 4.7% (internal) to 9.4% (external) fell on the background, so there was **no sign of a border shortcut**. Attention covered haemorrhages and the optic-disc region; the largest external error (proliferative predicted as no DR) was a dark, underexposed image with diffuse attention.

Meaning External errors are mostly systematic one-grade shifts, not random failures. Grad-CAM is coarse (16 × 16) and shows association, not causation; attention to the optic disc may reflect new vessels at the disc or merely its brightness, which cannot be distinguished without disc annotations.

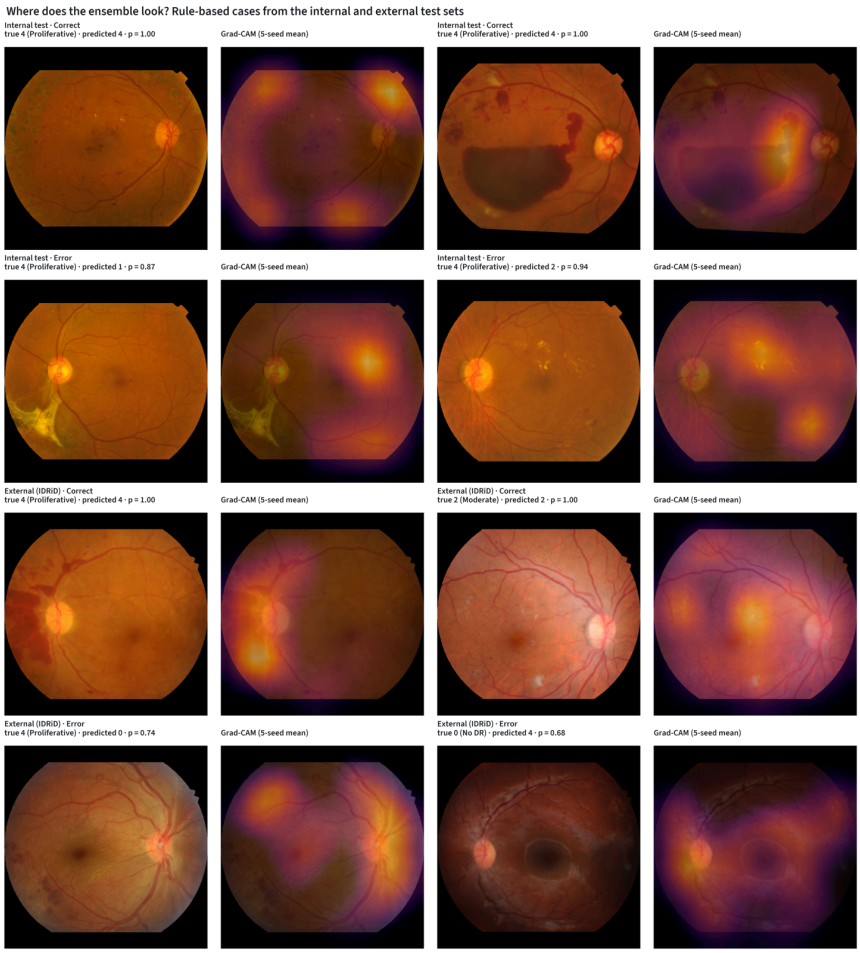


Figure 13. Grad-CAM (5-seed mean) for rule-selected cases: rows 1–2 internal test, rows 3–4 IDRiD; per set, the two most confident correct referable cases and the two largest errors.

4 Discussion

Answers to the research questions. (1) The data required substantial cleaning: about one in four APTOS duplicate groups carried contradictory grades, and the IDRiD archive was triplicated. (2) Measurements justified green-channel texture features, a validated mask fallback and anti-aliased resizing, and showed that CLAHE was unnecessary. (3) The experiment was protected against leakage and test-set tuning by data validation, a stratified split, a time-stamped protocol and one-time evaluations. (4) The CNN clearly outperformed hand-crafted radiomics. (5) Results were stable across seeds, but the model was fragile to noise and strong compression, in ways

SSIM did not predict. (6) The model did not generalise to IDRiD without recalibration; the drop was not due to case mix. (7) External errors were systematic one-grade shifts, with no evidence of border shortcuts.

Why the external drop matters. A validation QWK of 0.919 described performance on the source the model was built on. Internally the model missed only 1 of 64 severe or proliferative eyes; on IDRiD it missed 23 of 152 at the frozen threshold. Because proliferative DR needs urgent treatment, this is the most important clinical finding of the project. Plausible contributors are differences in camera optics and illumination (IDRiD is brighter, and brightening was harmful in the robustness tests), noise characteristics, and grading conventions, particularly for severe non-proliferative DR, which depends on subtle signs. These remain hypotheses. An earlier idea that IDRiD's larger downsampling factor removes more lesion detail is not supported, because both datasets are sampled identically relative to the retina (512 px across); only differences in field of view would change lesion size in the final images.

From research to clinical use. The first autonomous AI system for DR authorised by the U.S. FDA (IDx-DR, De Novo pathway, 2018) was supported by a prospective trial in primary-care clinics with pre-specified sensitivity and specificity endpoints [11]. Such a device needs defined indications for use, documented training data, non-clinical testing and evaluation on the intended population and equipment. This project provides retrospective internal evidence and one external evaluation; it shows precisely why site-specific recalibration and prospective validation would be required.

Limitations. No patient identifiers were available, so photographs of the same patient may appear across APTOS splits. IDRiD was a third-party repackaged archive, although the originals were recovered and matched the official counts. Grading conventions may differ between datasets, and reference labels are imperfect (as the conflicting duplicates show). A single architecture and fixed hyperparameters were used; training included no noise augmentation. Rare grades have few images, so their recall is uncertain. Grad-CAM is coarse. The recalibration analysis is post-hoc and would itself need independent external validation.

Future work. Train on several sources and cameras; add noise, blur and compression augmentation, motivated by the robustness results; use test-time augmentation; recalibrate on a local sample and validate on another; predict diabetic macular oedema; use lesion segmentation (for example a U-Net) to make the evidence for each grade explicit; and evaluate prospectively.

5 Conclusion

Careful data validation, measured preprocessing choices, a pre-registered evaluation strategy and seed ensembling produced a strong, stable and well-calibrated DR grading model on its development source (internal QWK 0.916, referable AUROC 0.984). The same model lost substantial accuracy on an external dataset from another camera and site (QWK 0.696), for reasons other than case mix, and missed some severe and proliferative eyes. The project's main lesson is that internal performance, however rigorously estimated, does not guarantee generalisation: robustness tests, external validation and local recalibration are required before any clinical use.

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Appendix A Course topics applied in the project

Course topic	Where it appears
Introduction to medical imaging	Fundus photography as an optical, non-ionising projection modality compared with X-ray, CT, MRI and PET (Section 1)
Digital images and representation	RGB arrays, bit depth, 17 resolutions, PNG vs JPEG, colour channels (2.1, 2.3)
Noise and image quality	Noise estimation, MSE, SSIM, task-specific quality (2.3, 3.2, 3.6)
Classical image processing	Histograms, Otsu, morphology (opening, closing, black-hat, top-hat), CLAHE, Gaussian and median filters (2.4, 3.2)
Sampling and filtering	Sampling theorem, anti-aliasing, measured aliasing, high-pass filtering and its edge effect (3.2)
Radiomics	GLCM/Haralick features, grey-level discretisation, intensity and morphological features; effect sizes vs p-values (2.6, 3.3)
Machine learning and generalisation	Cross-validation, regularisation (ridge as a prior), overfitting in learning curves, model selection on validation (2.5–2.6, 3.3)
CNNs and transfer learning	ResNet18, convolution, receptive fields, feature maps, ImageNet pretraining (2.6)
Evaluation and statistics	QWK, ROC/AUROC, sensitivity/specificity, bootstrap CIs, paired comparisons, calibration (2.7, 3.4)
Robustness and datasets	Controlled degradations, external validation, dataset validation (2.2, 3.6–3.7)
Regulation of AI devices	FDA authorisation of IDx-DR; requirements for clinical use (Section 4)
Assessing a research paper (IMRaD)	Structure of this report; patient selection and bias; robustness of the statistics

Appendix B Reproducibility

Order of decisions (time stamps from the project logs, UTC). Protocol created 22 Oct 2026, 22:32 → CLAHE tie-break rule recorded 23:53, before CNN-B was trained → configuration frozen 23 Oct, 00:30 → test reporting plan recorded 00:33 → internal test evaluated 11:35 → robustness protocol recorded 11:42 → IDRiD evaluated 12:10 → exploratory analyses 12:38.

Software. Python in Google Colab (NVIDIA T4): PyTorch 2.11, torchvision 0.26, scikit-learn 1.6.1, scikit-image 0.25.2, OpenCV 5.0, NumPy 2.1.3, pandas 2.2.3, SciPy 1.16.3, Matplotlib 3.10, ImageHash 4.3.2. Seeds: 42 for all data operations; 42, 1, 2, 3, 4 for the CNN ensemble.

Code and outputs. Shared modules (*preprocessing.py, evaluation.py, cnn.py, robustness.py, plotting.py*) are used by every notebook cell, so the same functions compute all metrics. Splits, protocol, configurations, training logs, predictions, every table and every figure are saved and can be regenerated from the notebook; raw images and model weights are kept out of the shared package.